# Chapter 0
# Introduction

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- Course syllabus
- A brief introduction to stochastic processes

# People

- **Instructor:**

  Tiefeng Jiang, jiang040@cuhk.edu.cn
  Office: DY 610
  Office Hour: 3 - 4 pm, Monday and Wednesday (from week 2)

- **TA:**
  Yida Wu, 224040333@link.cuhk.edu.cn OH: Tue 14:00-15:00, TxC 716
  Surui Wang, 224040338@link.cuhk.edu.cn OH: Thur 16:30-17:30, TA105
  Yuchen Yang, 224040354@link.cuhk.edu.cn OH: Wed 16:00-17:00, THL311
  Zheyuan Lin, 223040225@link.cuhk.edu.cn OH: Wed 13:00-14:00, TxC604, seat15

- Relationship to other sessions:
  - Covering same topics, same homework and exam questions
  - Sharing tutorial sessions and similar lecture delivery

# Course objective and topics

- Introduction to basic stochastic process models, theory and some applications.

- Stochastic Processes
  1. Discrete time Markov chain
  2. Continuous time Markov chain
  3. Brownian motion

- Applications
  1. Queueing systems: hospital, telecommunication networks
  2. Reinforcement learning: intervention in healthcare

# Assessment and Course Materials

- Homework and quizzes (30%)
- Midterm (30%)
- Final (40%)
  - Closed-book

- Course Materials
  - **Textbook**(required): *Introduction to Probability Models*, by Sheldon M. Ross, 10th edition, Elsevier, ISBN: 978-0-12-375686-2
  - **Slides** & **Notes**

# Lecture and Assignments

- Homework will be assigned in class every other week and due back in two weeks through BB.
- You need to write down detailed steps of computation for each question.
- Unless you have very very very special reason, late submission of assignment will get no credit.
- Partially or wholly copied assignments will be penalized and/or reported as plagiarism. We will not tolerate students who cheat.
- Although the lectures will be recorded, please join the on-site lecture if you are on campus.

# Topics

1. Conditional Probability and Expectation
2. Random walks
3. Discrete time Markov chains
4. Poisson processes
5. Continuous time Markov chains
6. Special topic I: queueing models and applications
7. Special topic II: stochastic process and machine learning*
8. Martingales and Brownian motion.*

* subject to changes according to the progress.

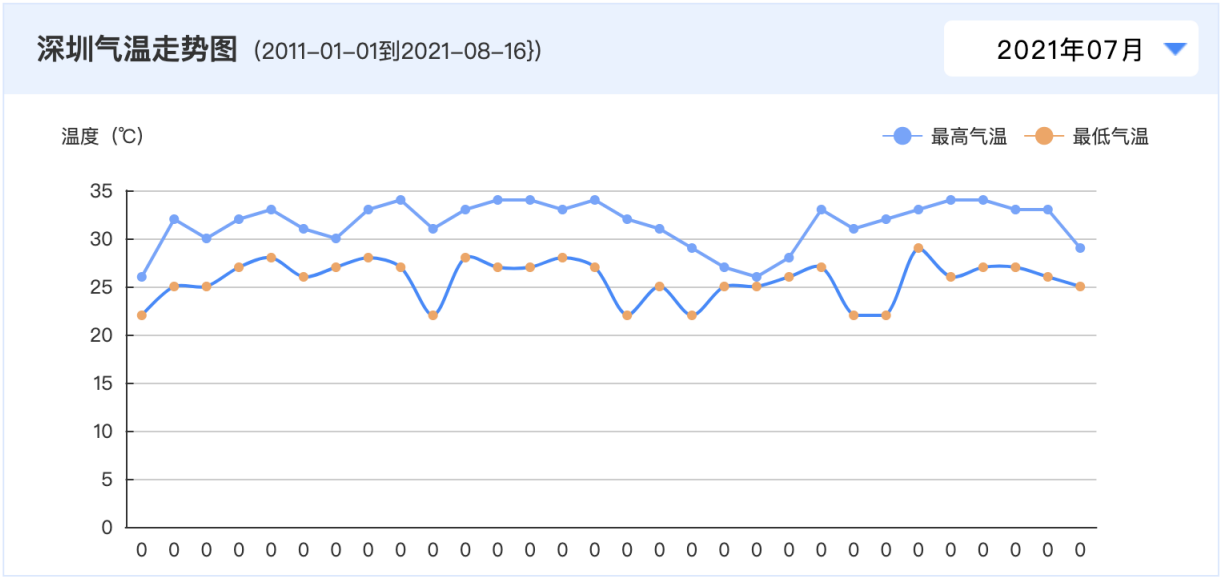
# My expectation on you

- I will treat you as undergraduate students with scientific curiosity, good planning and independent-study ability.
- I hope you could understand the high-level ideas, master the mathematics and also form a good intuition on basic stochastic processes and analysis methods through this course.
- If you have any difficulty with the course materials, you are always welcome to talk to me and we can see how your problem can be resolved.

# Introduction

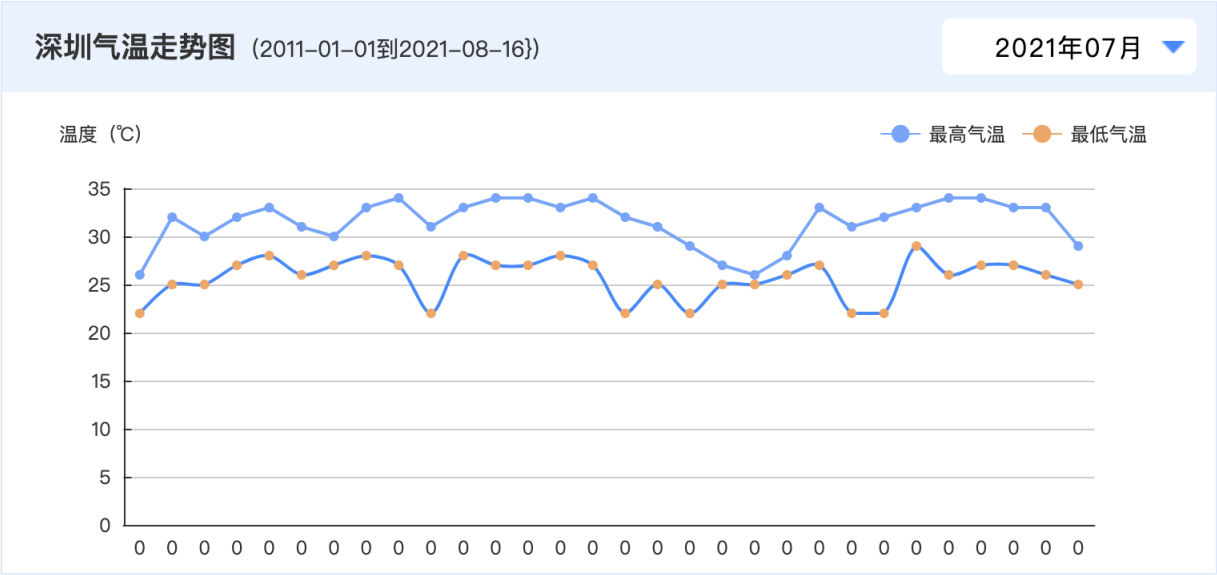
- A stochastic process $\{X(t), t \in T\}$ is a collection of random variables, that is, for each $t \in T$, $X(t)$ is a random variable.

- The set $T$ is called the index set of the process, and a $t \in T$ is called an index.
  - Often, $t$ is referred to time.
  - We call $X(t)$ the state of the process at time $t$.



# Introduction

- If $T$ is a *countable* set, such as $T = \{1, 2, 3, ...\}$, $\{X(t)\}$ is called a discrete-time stochastic process.
- If $T$ is an interval of the real line, such as $T = \{t : t \geq 0\}$, $\{X(t)\}$ is called a continuous time stochastic process.
- The state space of a stochastic process is defined as the set of all possible values that $X(t)$ can assume (for all $t \in T$).



# Examples

- What are the index set and state space of the stochastic process?
  - Outcomes of flipping a coin
  - Number of students in Pandora from 10 am to 2 pm

- Any examples you can come up with from daily life or previous courses?

# A toy example

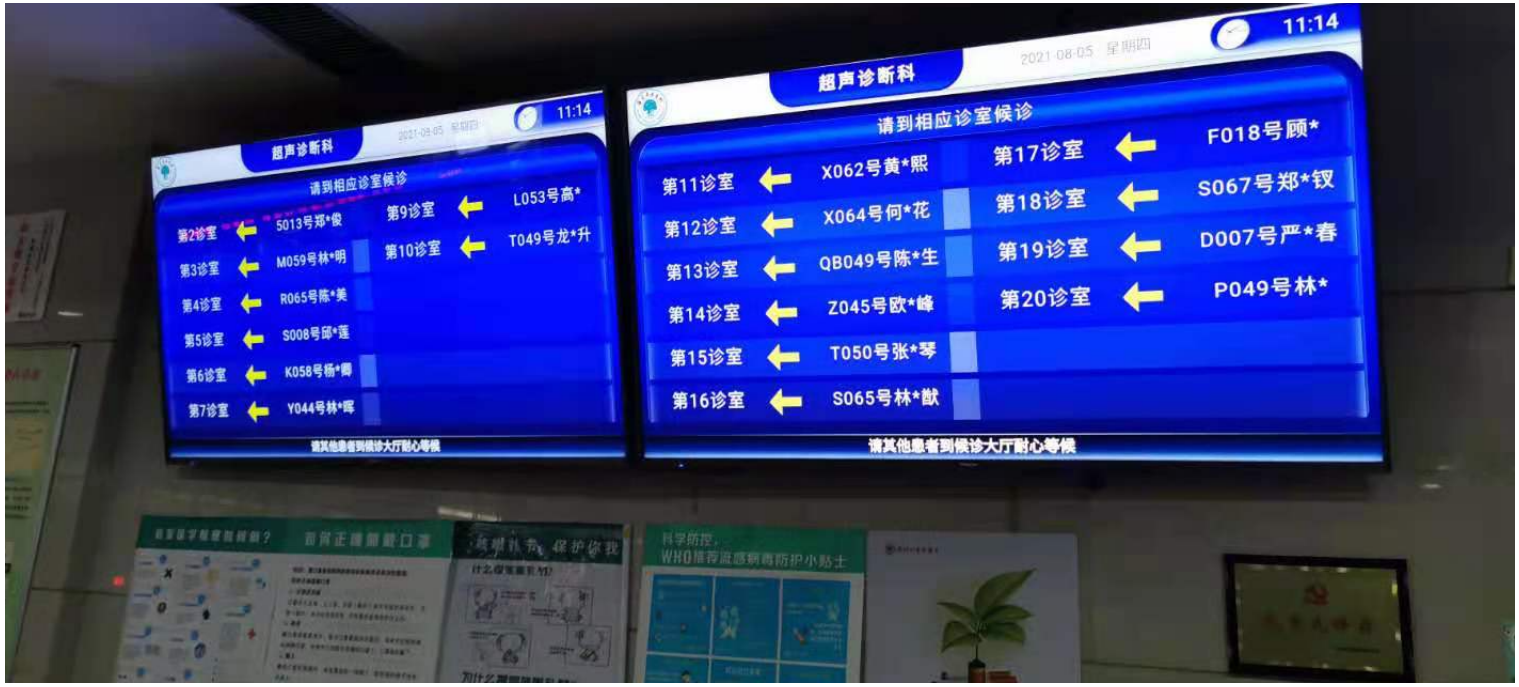
- **(Gambler's ruin problem)** Suppose a gambler has 1 dollar in pocket at time 0. Each time, he is equially likely to win or lose 1 dollar. What is the probability that the gambler wins $n$ dollars before ruin (lose all his money).
- How to build a stochastic process to describe the gambler's wealth?
  - State space.
  - Transition distribution.

# A toy example

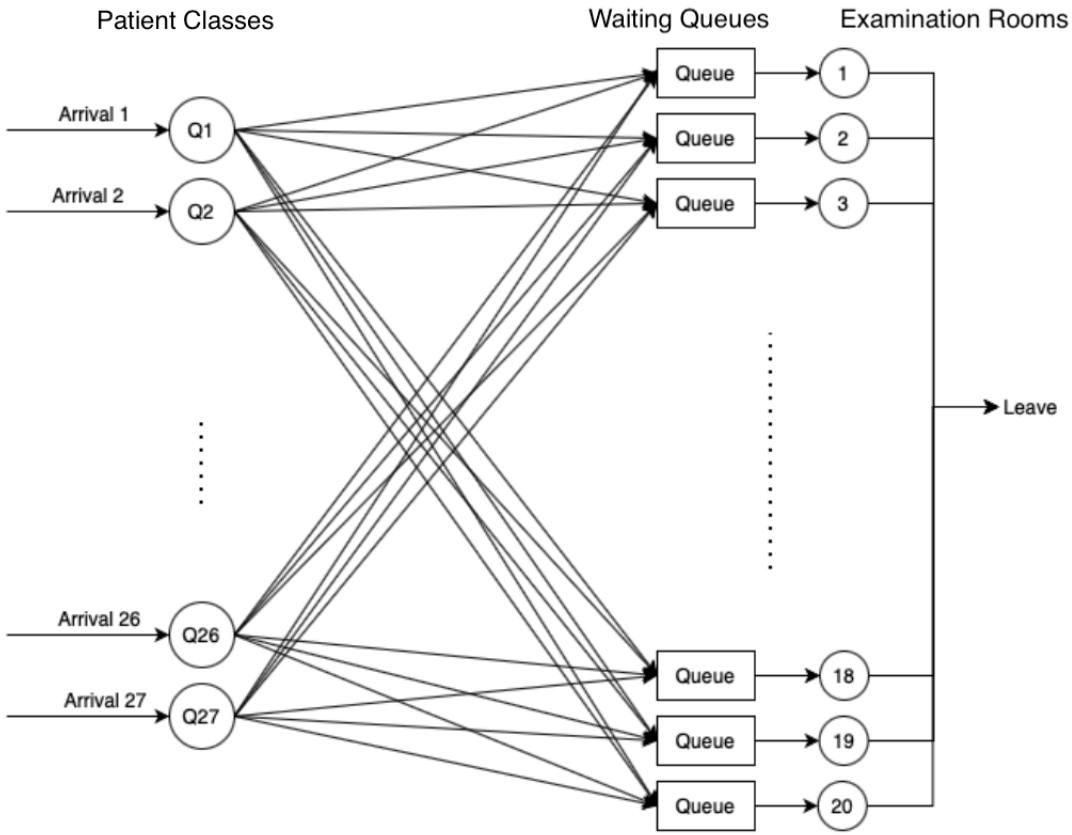
- A generalization of this simple model can be used to compute the insurance company's ruin probability, which is a fundamental problem in actuarial science.

# A Real Stochastic System

- **Goal:** Evaluate the performance of the ultrasound center in a large hospital in Fujian Providence of China.
- We shall measure system performance by the length of waiting queues.



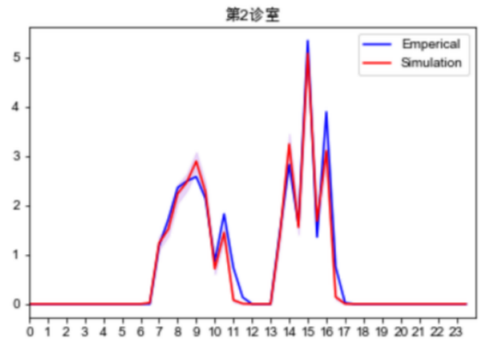
# A Real Stochastic System



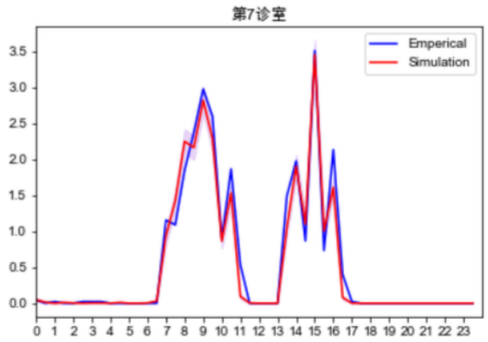
# A Real Stochastic System

- Continuous Time.
- Multi-dimension state vector:
  - State of the examination rooms: busy or idle
  - Examination times of patients in the examination rooms
  - Number of patients in the queues.
  - Epoch between two consequence arrivals in the same patient class
- How to specify the distribution of system dynamics?
  - Arrival process (e.g. Poisson process)
  - Examination time sequence (e.g. i.i.d. exponential)
  - Routing probabilities

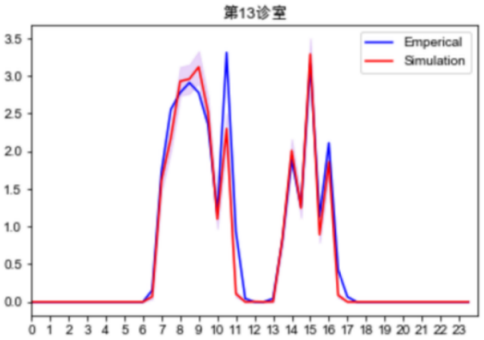
# Simulator v.s. Real System



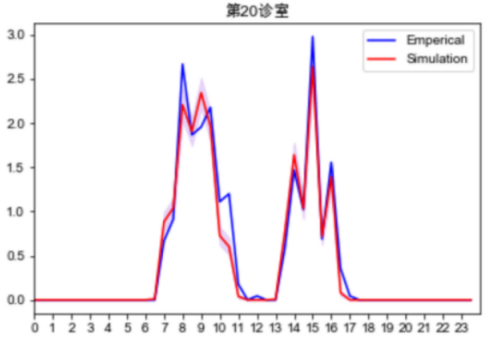

(a) 2-nd examination room




(b) 7-th examination room




(c) 13-th examination room




(d) 20-th examination room

# Basic stochastic processes

- The definition of stochastic processes is very general. The property of an arbitrary stochastic process can be very complicated. So people start to study some special classes of stochastic processes that are mathematically attractive. Those processes are building blocks when constructing stochastic models in practical applications.
- In this course, we will study the following types of stochastic processes:
  - random walk
  - discrete-time Markov chain
  - Poisson (counting) process
  - continuous-time Markov chain
  - Brownian motion

# Review of Probability

- Counting
- Probability
- Discrete distributions
- Continuous distributions
- Multivariate distributions
- Conditional distributions

( Chapter 1 to Chapter 5 of Textbook.)

# Counting

- Addition and Multiplication principles
- Selection of $r$ from $n$ distinct objects, with replacement and without replacement
- Arrangement of $n$ objects with $r$ distinct types.
- Partition of $n$ indistinguishable objects into $r$ distinct groups where some groups can be empty.

# Mathematics of Probability

- Kolmogorov's Axiom
- Inclusion-Exclusion Principle
- Conditional Probability
- Law of Total Probability
- Bayes' Theorem (Bayes' Rule, Bayes' Law)

# Probability Distribution

- Definition of probability mass function (pmf)
- Definition of probability density function (pdf)
- Properties of pmf / pdf
- Expectation
- Mean and Variance
- Moment and Moment Generating Function

# Discrete Distributions

- Discrete uniform
- Bernoulli
- Binomial
- Geometric
- Poisson

# Continuous Distributions

- Uniform
- Beta
- Exponential
- Gamma
- Normal

# Multivariate distributions

- Joint and marginal distributions
- Independence of random variables
- Covariance
- Correlation coefficient
- Sum of independence random variables

# Conditional distributions

- Conditional distribution function of $Y$ given $X = x$
- Conditional expectation of $g(Y)$ given $X = x$
- Conditional mean of $Y$ given $X = x$
- Conditional variance of $Y$ given $X = x$

# Limit Theorems

- Law of Large Number (LLN)
- Central Limit Theorem (CLT)